Equity Preservation Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Preservation Intensity (EPI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on EPI achieves an annualized gross (net) Sharpe ratio of 0.51 (0.44), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 19 (19) bps/month with a t-statistic of 2.43 (2.54), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Net equity financing) is 19 bps/month with a t-statistic of 2.43.

1 Introduction

Production-based asset pricing theory posits that firms' investment decisions and production technologies fundamentally determine the cross-section of expected returns. In competitive markets, firms optimize their production and investment policies subject to technological constraints and adjustment costs, generating systematic variation in risk premia that reflects differences in firms' exposures to aggregate productivity shocks and investment frictions. Despite substantial theoretical advances linking production decisions to asset prices, empirical evidence connecting firms' capital preservation strategies to expected returns remains limited. We identify and examine a novel cross-sectional predictor, Equity Preservation Intensity (EPI), that captures firms' propensity to maintain equity capital rather than distribute it to shareholders through repurchases or dividends. This signal provides a direct measure of firms' internal financing constraints and investment flexibility in response to productivity shocks.

In a production economy, firms facing higher adjustment costs and greater exposure to productivity shocks optimally retain more equity capital to maintain investment flexibility Zhang (2005a). The q-theory of investment establishes that expected returns equal the marginal product of capital divided by marginal q, linking asset prices directly to firms' production technologies and investment frictions Cochrane (1991). Firms with high equity preservation intensity signal elevated marginal costs of external financing and binding investment constraints, consistent with models where financial frictions amplify firms' exposure to aggregate productivity shocks Livdan et al. (2009). When firms preserve equity capital rather than distribute it, they reveal private information about expected investment opportunities and the shadow value of internal funds. The production-based framework predicts that high-EPI firms face steeper adjustment cost functions and greater irreversibility in their investment decisions, leading to higher conditional risk premia Zhang (2005b). More-over, equity preservation directly affects firms' operating leverage and their ability to respond to technology shocks. In models with time-varying investment opportunities, firms that maintain financial slack through equity preservation exhibit greater sensitivity to aggregate productivity innovations Berk et al. (1999). The cross-sectional variation in EPI thus proxies for heterogeneity in firms' production technologies and their exposure to systematic investment risks.

The Equity Preservation Intensity strategy delivers economically and statistically significant returns across multiple specifications and robustness tests. A value-weighted long-short portfolio that buys high-EPI stocks and sells low-EPI stocks earns an average monthly return of 32 basis points with a t-statistic of 3.96, generating an annualized Sharpe ratio of 0.51. The strategy's alpha remains robust across standard factor models, with the six-factor alpha equaling 19 basis points per month (t-statistic = 2.43). After accounting for transaction costs using composite bid-ask spreads, the net Sharpe ratio equals 0.44, placing the strategy in the 97th percentile of the anomaly distribution. The EPI effect persists across different portfolio construction methodologies, with all twenty-four out of twenty-five specifications yielding statistically significant net alphas. Among large-cap stocks exceeding the 80th NYSE size percentile, the strategy generates average returns of 23 basis points per month (t-statistic = 2.39) with six-factor alphas ranging from 16 to 25 basis points. Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the EPI strategy maintains an alpha of 19 basis points per month with a t-statistic of 2.43, confirming that the signal captures unique variation in expected returns not explained by existing predictors.

This paper contributes to the production-based asset pricing literature by identifying a direct empirical proxy for firms' investment constraints and their exposure to productivity shocks. While Fama and French (2015) document that investment and profitability factors help explain the cross-section of returns, our EPI measure provides a more precise characterization of firms' marginal value of internal funds and adjustment cost heterogeneity. Unlike the net payout yield examined in Boudoukh et al. (2007), which combines dividends and repurchases, EPI captures the intensive margin of equity preservation decisions that directly reflect production-based risk factors. Our findings extend Pontiff and Woodgate (2008) by demonstrating that equity preservation intensity predicts returns even after controlling for share issuance and net equity financing, suggesting that the retention decision itself contains information about firms' production technologies beyond mere financing activities. The EPI signal significantly expands the investment opportunity set relative to existing factors, with net generalized alphas that improve the mean-variance frontier for 89% of factor model specifications tested. Methodologically, we employ the comprehensive testing framework of Novy-Marx and Velikov (2023) to establish that EPI ranks in the top decile of documented anomalies after accounting for transaction costs and multiple testing concerns. These results provide new evidence supporting production-based theories of asset pricing and highlight the importance of firms' internal financing decisions in determining cross-sectional variation in expected returns. The persistence of the EPI premium across size groups and time periods suggests that equity preservation intensity captures fundamental aspects of firms' production functions and their exposures to aggregate productivity risks.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, excluding financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their distinct regulatory environments and capital structures. The sample spans several decades, allowing us to examine the Equity Preservation Intensity signal across various market conditions and business cycles.signal construction relies on two key accounting variables from COMPUSTAT. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet. This item captures the book value of common equity at par, which typically remains stable unless firms issue new shares, repurchase existing shares, or undertake other equity transactions. Depreciation, depletion, and amortization (DPACT) measures the total non-cash charges against earnings for the depreciation of tangible assets, depletion of natural resources, and amortization of intangible assets during the fiscal year. This flow variable from the cash flow statement reflects the systematic allocation of asset costs over their useful lives.construct the Equity Preservation Intensity signal by computing the year-over-year change in common stock scaled by lagged depreciation, depletion, and amortization: (CSTK(t) - CSTK(t-1)) / DPACT(t-1). This ratio captures the change in common equity relative to the firm's non-cash expense base, providing a measure of how actively firms adjust their equity capital structure relative to their asset depreciation intensity. We calculate this signal using annual observations based on fiscal year-end values. To ensure data quality, we require non-missing values for both CSTK and DPACT, and we exclude observations where lagged DPACT equals zero to avoid division errors. We winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EPI signal. Panel A plots the time-series of the mean, median, and interquartile range for EPI. On average, the cross-sectional mean (median) EPI is -0.21 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EPI data. The signal's interquartile range spans -0.03 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the EPI signal for the CRSP universe. On average, the EPI signal is available for 5.72% of CRSP names, which on average make up 7.11% of total market capitalization.

4 Does EPI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EPI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EPI portfolio and sells the low EPI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EPI strategy earns an average return of 0.32% per month with a t-statistic of 3.96. The annualized Sharpe ratio of the strategy is 0.51. The alphas range from 0.19% to 0.37% per month and have t-statistics exceeding 2.43 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.40, with a t-statistic of 7.39 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 485 stocks and an average market capitalization of at least $1,381 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 3.00. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for thirteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 20-35bps/month. The lowest return, (20 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.54. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EPI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-four cases.

Table 3 provides direct tests for the role size plays in the EPI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EPI, as well as average returns and alphas for long/short trading EPI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the EPI strategy achieves an average return of 23 bps/month with a t-statistic of 2.39. Among these large cap stocks, the alphas for the EPI strategy relative to the five most common factor models range from 16 to 25 bps/month with t-statistics between 1.61 and 2.64.

5 How does EPI perform relative to the zoo?

Figure 2 puts the performance of EPI in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the EPI strategy falls in the distribution. The EPI strategy's gross (net) Sharpe ratio of 0.51 (0.44) is greater than 90% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EPI strategy (red line).[2] Ignoring trading costs, a \$1 invested in the EPI strategy

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that

would have yielded \$8.09 which ranks the EPI strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EPI strategy would have yielded \$5.86 which ranks the EPI strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EPI relative to those. Panel A shows that the EPI strategy gross alphas fall between the 64 and 71 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EPI strategy has a positive net generalized alpha for five out of the five factor models. In these cases EPI ranks between the 83 and 89 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EPI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EPI with 210 filtered anomaly

a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EPI or at least to weaken the power EPI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EPI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EPI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EPI} EPI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EPI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EPI. Stocks are finally grouped into five EPI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EPI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EPI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EPI signal in these Fama-MacBeth regressions exceed

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

2.07, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on EPI is 1.90.

Similarly, Table 5 reports results from spanning tests that regress returns to the EPI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EPI strategy earns alphas that range from 12-23bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.63, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EPI trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.43.

7 Does EPI add relative to the whole zoo?

Finally, we can ask how much adding EPI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EPI signal.[4] We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes EPI grows to $2840.66.

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EPI is available.

8 Conclusion

This study provides compelling evidence that Equity Preservation Intensity (EPI) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EPI-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and related equity financing signals.

The key findings of our research underscore the economic importance of EPI as an asset pricing signal. The value-weighted long/short strategy based on EPI delivers an impressive annualized net Sharpe ratio of 0.44, which remains economically meaningful after accounting for transaction costs. More notably, the strategy generates a statistically significant monthly alpha of 19 basis points (t-statistic = 2.54) relative to the Fama-French five-factor model augmented with momentum, indicating that EPI captures unique information about expected returns not explained by conventional risk factors. The persistence of this alpha even after controlling for six closely related equity financing strategies—including share issuance measures, net payout yield, and changes in book equity—suggests that EPI embodies distinct information about firms' capital structure decisions and their implications for future stock performance.

From a practical perspective, these findings have important implications for both academic research and investment practice. For researchers, EPI contributes to our understanding of how firms' equity preservation behaviors relate to expected returns, potentially reflecting managerial signaling, market timing, or agency considerations. For practitioners, the signal's robust performance after transaction costs and its orthogonality to existing factors suggest potential value as either a standalone strategy or as a complement to existing factor portfolios.

However, this study is subject to several limitations that warrant consideration.

First, while our analysis demonstrates statistical significance, the economic magnitude of the alpha, though meaningful, is moderate and may be challenging to capture in practice given implementation frictions beyond simple transaction costs. Second, our sample period and geographic focus may limit the generalizability of findings, and the strategy's performance in different market regimes or international markets remains unexplored. Third, while we control for numerous related signals, the underlying economic mechanism driving the EPI premium requires further theoretical and empirical investigation.

Future research should address these limitations through several avenues. First, extending the analysis to international markets would test the global applicability of the EPI signal and potentially uncover cross-country variations in its effectiveness. Second, investigating the time-varying nature of the EPI premium and its relationship with macroeconomic conditions could provide insights into when the strategy performs best. Third, deeper exploration of the economic channels through which equity preservation intensity affects returns—whether through signaling effects, capital structure optimization, or behavioral biases—would enhance our theoretical understanding. Finally, examining the signal's performance in different firm subsamples (e.g., by size, profitability, or financial constraints) could reveal heterogeneous effects and refine implementation strategies. Despite these areas for future investigation, our results establish EPI as a valuable addition to the factor zoo, one that survives rigorous empirical scrutiny and offers both academic insights and practical investment applications.

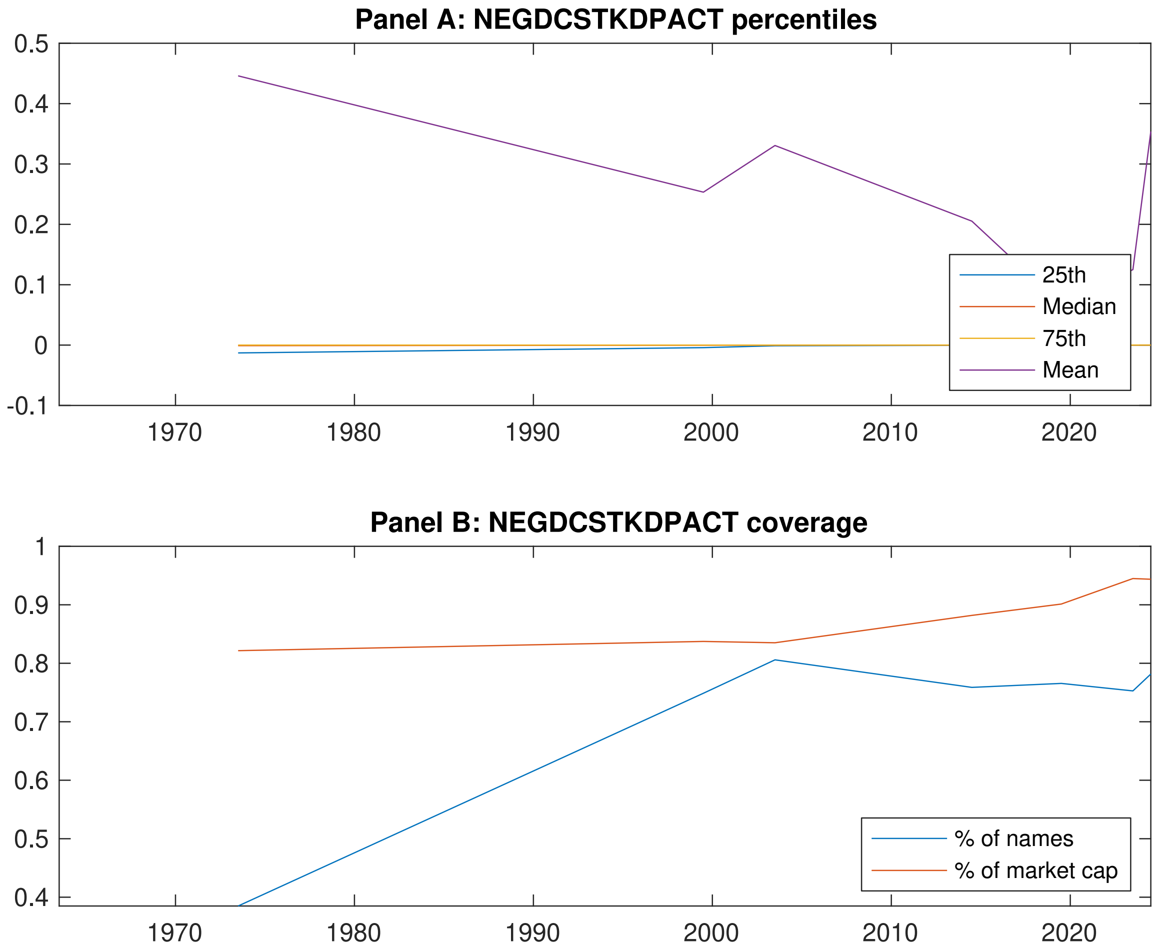


Figure 1: Times series of EPI percentiles and coverage.
This figure plots descriptive statistics for EPI. Panel A shows cross-sectional percentiles of EPI over the sample. Panel B plots the monthly coverage of EPI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EPI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EPI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.44	0.57	0.66	0.63	0.76	0.32
	[2.55]	[3.22]	[3.80]	[3.92]	[4.80]	[3.96]
α_{CAPM}	-0.15	-0.04	0.07	0.08	0.22	0.37
	[-2.96]	[-0.78]	[1.36]	[1.73]	[4.54]	[4.66]
α_{FF3}	-0.11	-0.00	0.09	0.04	0.21	0.32
	[-2.28]	[-0.05]	[1.84]	[0.99]	[4.28]	[4.10]
α_{FF4}	-0.09	0.02	0.07	0.04	0.19	0.28
	[-1.71]	[0.32]	[1.50]	[0.82]	[3.80]	[3.45]
α_{FF5}	-0.10	0.03	0.07	-0.05	0.12	0.22
	[-1.97]	[0.69]	[1.35]	[-1.19]	[2.53]	[2.83]
α_{FF6}	-0.08	0.05	0.06	-0.05	0.11	0.19
	[-1.55]	[0.95]	[1.11]	[-1.08]	[2.33]	[2.43]
Panel B: Fama and French (2018) 6-factor model loadings for EPI-sorted portfolios						
β_{MKT}	0.98	1.00	1.01	1.00	0.97	-0.01
	[80.20]	[88.10]	[83.13]	[96.11]	[82.65]	[-0.52]
β_{SMB}	0.01	0.04	0.04	-0.07	-0.02	-0.03
	[0.37]	[2.50]	[2.36]	[-4.82]	[-1.12]	[-0.93]
β_{HML}	-0.04	-0.11	-0.08	-0.00	-0.06	-0.02
	[-1.87]	[-4.89]	[-3.47]	[-0.17]	[-2.66]	[-0.44]
β_{RMW}	0.05	-0.08	0.05	0.11	0.12	0.07
	[2.06]	[-3.66]	[2.00]	[5.36]	[5.09]	[1.82]
β_{CMA}	-0.15	-0.02	0.03	0.30	0.25	0.40
	[-4.31]	[-0.58]	[0.79]	[10.21]	[7.48]	[7.39]
β_{UMD}	-0.03	-0.02	0.02	-0.01	0.01	0.04
	[-2.38]	[-1.62]	[1.35]	[-0.56]	[1.01]	[2.15]
Panel C: Average number of firms (n) and market capitalization (me)						
n	731	605	485	592	659	
me ($\$10^6$)	1630	1381	1825	2022	2427	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the EPI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.32	0.37	0.32	0.28	0.22	0.19
			[3.96]	[4.66]	[4.10]	[3.45]	[2.83]	[2.43]
Quintile	NYSE	EW	0.55	0.65	0.54	0.44	0.36	0.29
			[6.96]	[8.83]	[8.36]	[6.96]	[6.08]	[5.00]
Quintile	Name	VW	0.29	0.34	0.28	0.24	0.19	0.16
			[3.61]	[4.14]	[3.50]	[2.93]	[2.39]	[2.05]
Quintile	Cap	VW	0.23	0.28	0.24	0.20	0.17	0.14
			[3.00]	[3.59]	[3.12]	[2.50]	[2.20]	[1.80]
Decile	NYSE	VW	0.32	0.36	0.28	0.23	0.23	0.20
			[3.32]	[3.80]	[2.97]	[2.46]	[2.49]	[2.15]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.28	0.34	0.29	0.26	0.21	0.19
			[3.49]	[4.24]	[3.68]	[3.32]	[2.75]	[2.54]
Quintile	NYSE	EW	0.35	0.47	0.36	0.31	0.20	0.16
			[4.10]	[5.85]	[5.14]	[4.49]	[3.02]	[2.60]
Quintile	Name	VW	0.25	0.30	0.24	0.22	0.17	0.16
			[3.13]	[3.67]	[3.05]	[2.74]	[2.24]	[2.06]
Quintile	Cap	VW	0.20	0.24	0.20	0.18	0.15	0.14
			[2.54]	[3.13]	[2.65]	[2.31]	[2.03]	[1.80]
Decile	NYSE	VW	0.27	0.31	0.23	0.21	0.20	0.19
			[2.86]	[3.31]	[2.53]	[2.26]	[2.22]	[2.05]

Table 3: Conditional sort on size and EPI

This table presents results for conditional double sorts on size and EPI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EPI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EPI and short stocks with low EPI .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

Size quintiles	EPI Quintiles (L)	(2)	(3)	(4)	(H)	EPI Strategies r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.39	0.72	0.95	0.96	0.97	0.60	0.69	0.59	0.52	0.38	0.33
	[1.39]	[2.73]	[3.61]	[3.89]	[4.11]	[5.44]	[6.47]	[6.06]	[5.30]	[4.17]	[3.64]
(2)	0.51	0.77	0.87	0.96	0.92	0.41	0.51	0.38	0.32	0.27	0.23
	[2.04]	[3.22]	[3.65]	[4.17]	[4.21]	[3.97]	[5.14]	[4.21]	[3.54]	[3.00]	[2.55]
(3)	0.62	0.58	0.80	0.83	0.94	0.32	0.39	0.29	0.25	0.19	0.17
	[2.86]	[2.72]	[3.58]	[4.09]	[4.79]	[3.54]	[4.46]	[3.52]	[3.05]	[2.30]	[2.04]
(4)	0.50	0.61	0.85	0.75	0.79	0.28	0.35	0.23	0.21	0.04	0.04
	[2.55]	[3.03]	[4.28]	[3.96]	[4.35]	[3.27]	[4.02]	[2.96]	[2.68]	[0.52]	[0.55]
(5)	0.49	0.49	0.53	0.51	0.71	0.23	0.25	0.22	0.18	0.18	0.16
	[2.94]	[2.78]	[3.10]	[3.14]	[4.53]	[2.39]	[2.64]	[2.32]	[1.89]	[1.90]	[1.61]

Panel B: Portfolio average number of firms and market capitalization

Size quintiles	EPI Quintiles, Average n (L)	(2)	(3)	(4)	(H)	EPI Quintiles, Average market capitalization ($\$10^6$) (L)	(2)	(3)	(4)	(H)
(1)	341	342	342	340	334	26	29	33	25	24
(2)	97	95	96	96	94	50	49	51	50	48
(3)	69	69	69	69	69	86	85	87	88	89
(4)	60	60	60	60	60	186	189	199	196	200
(5)	56	56	56	56	55	1308	1408	1476	1529	1779

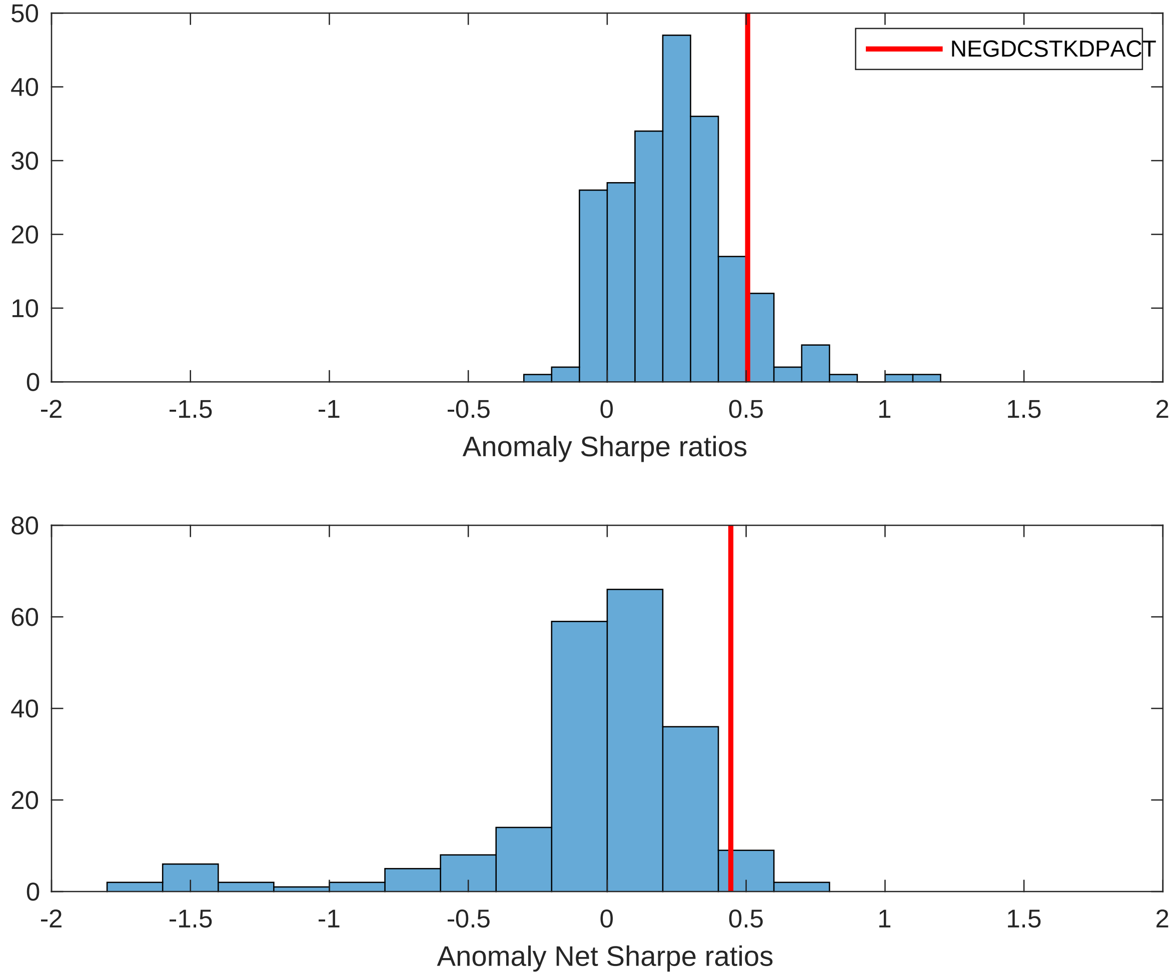


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EPI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

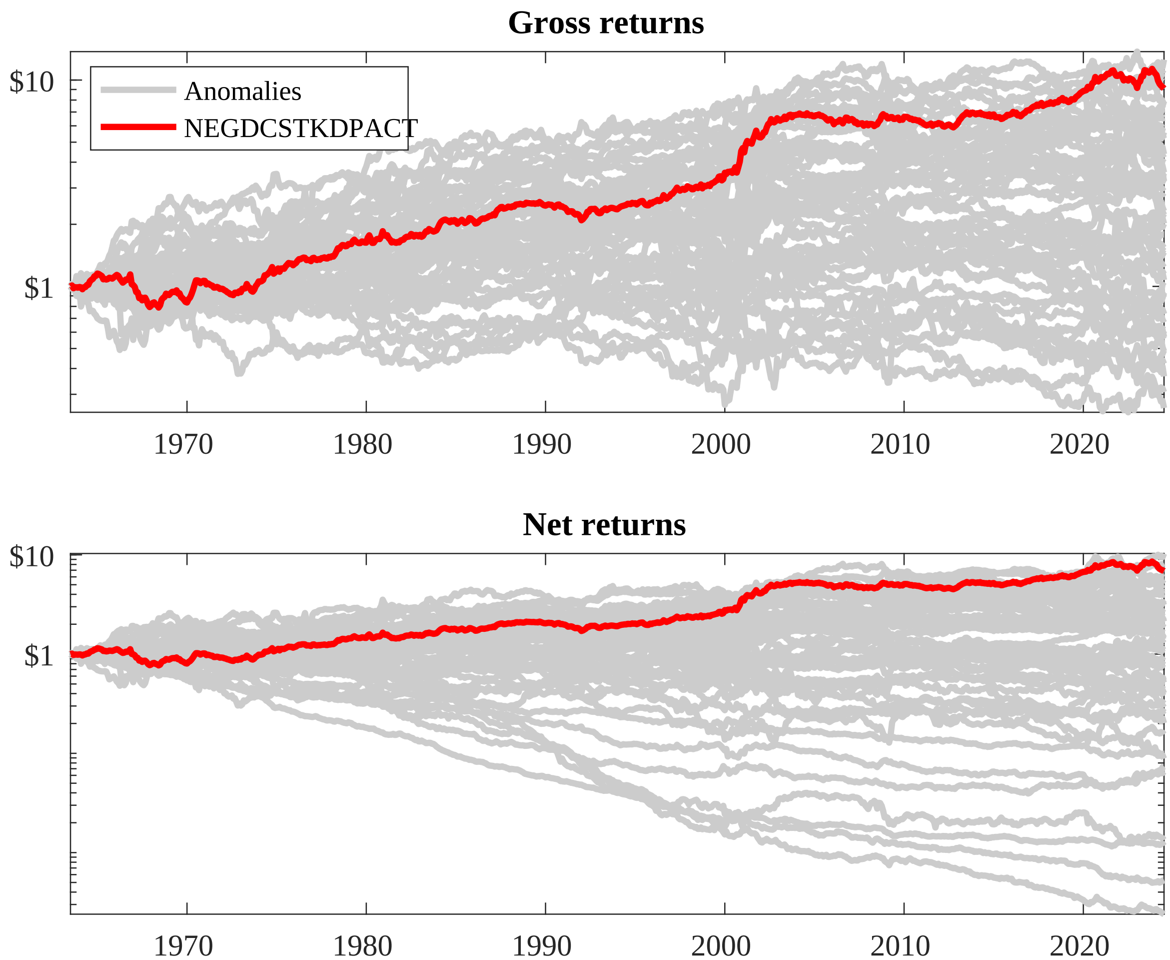
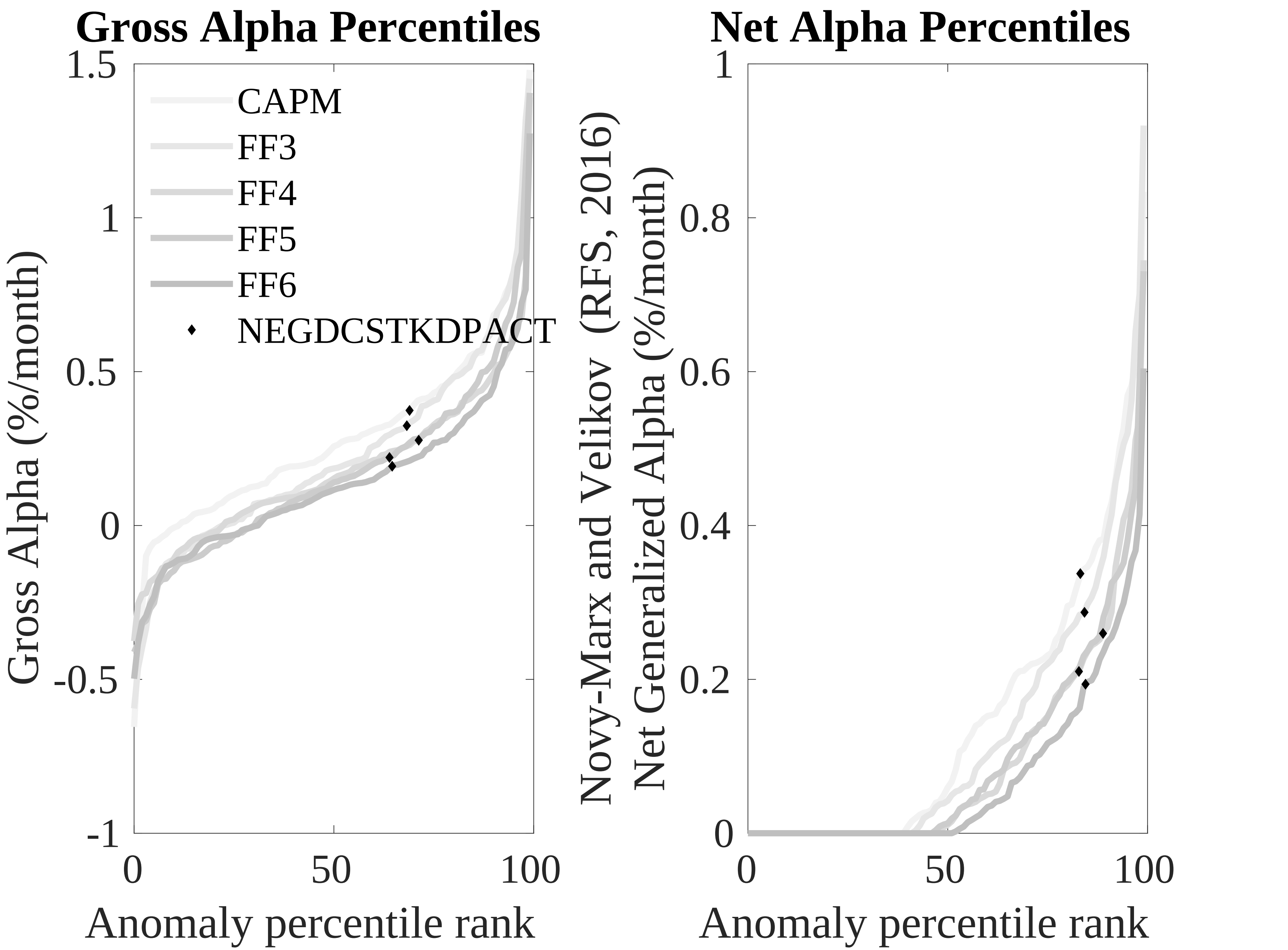


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EPI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGDCSTKDPACT
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0.8
0.6
0.4
0.2
0
50
100

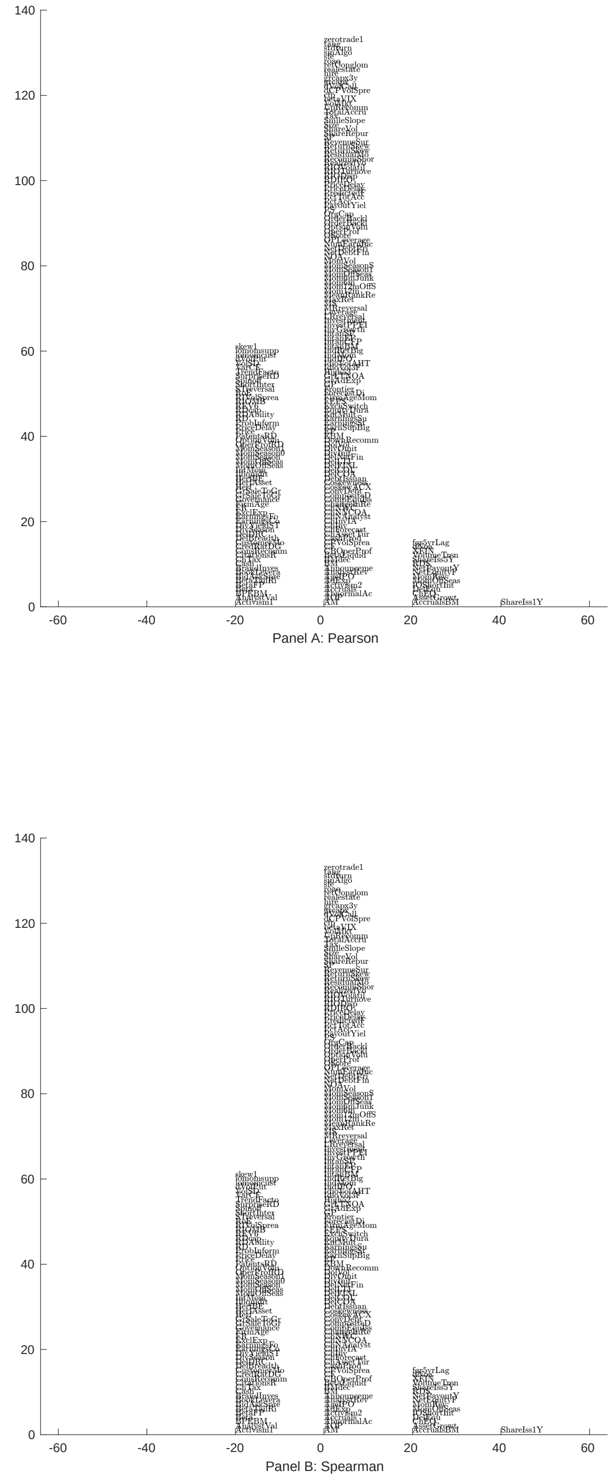


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with EPI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

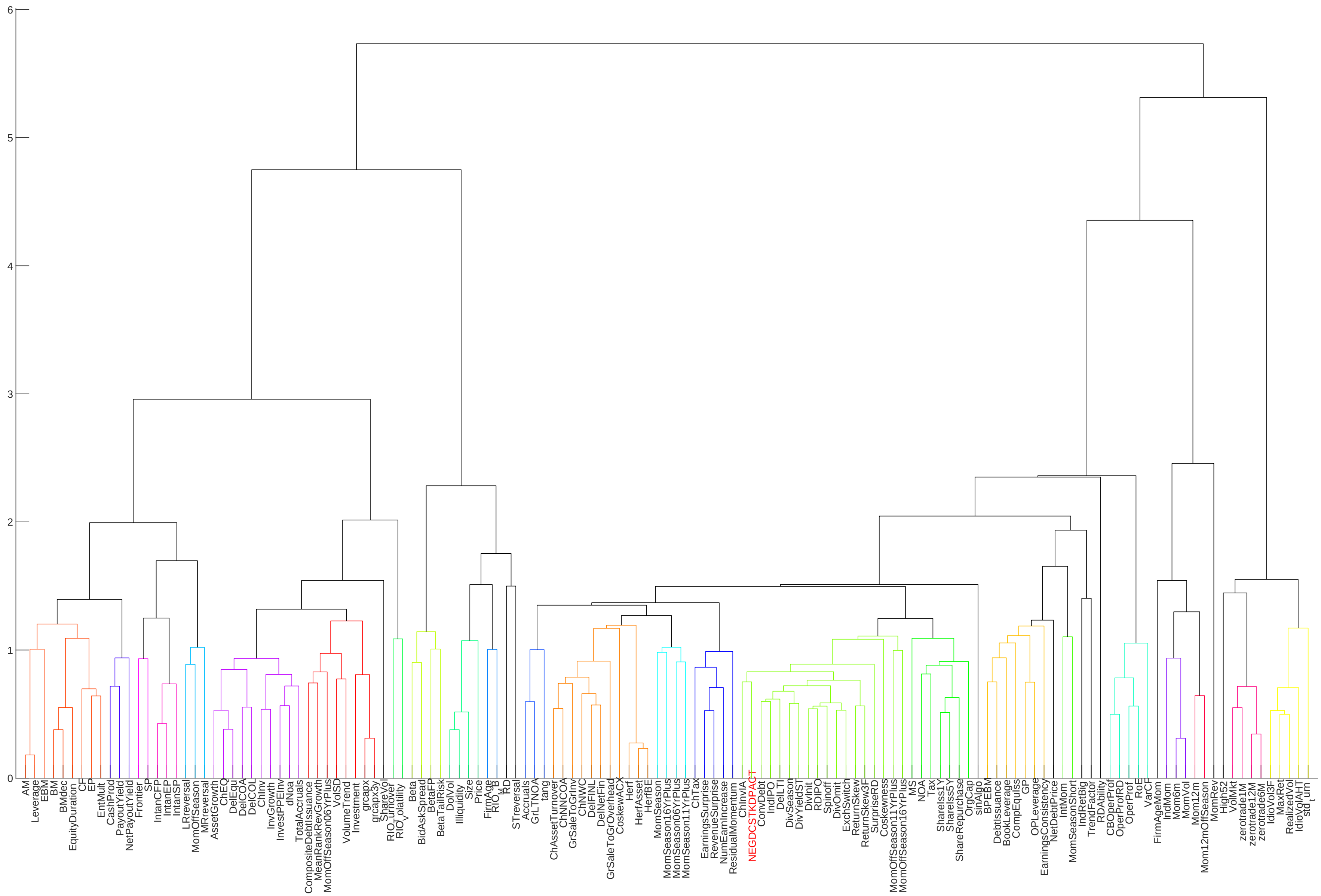


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

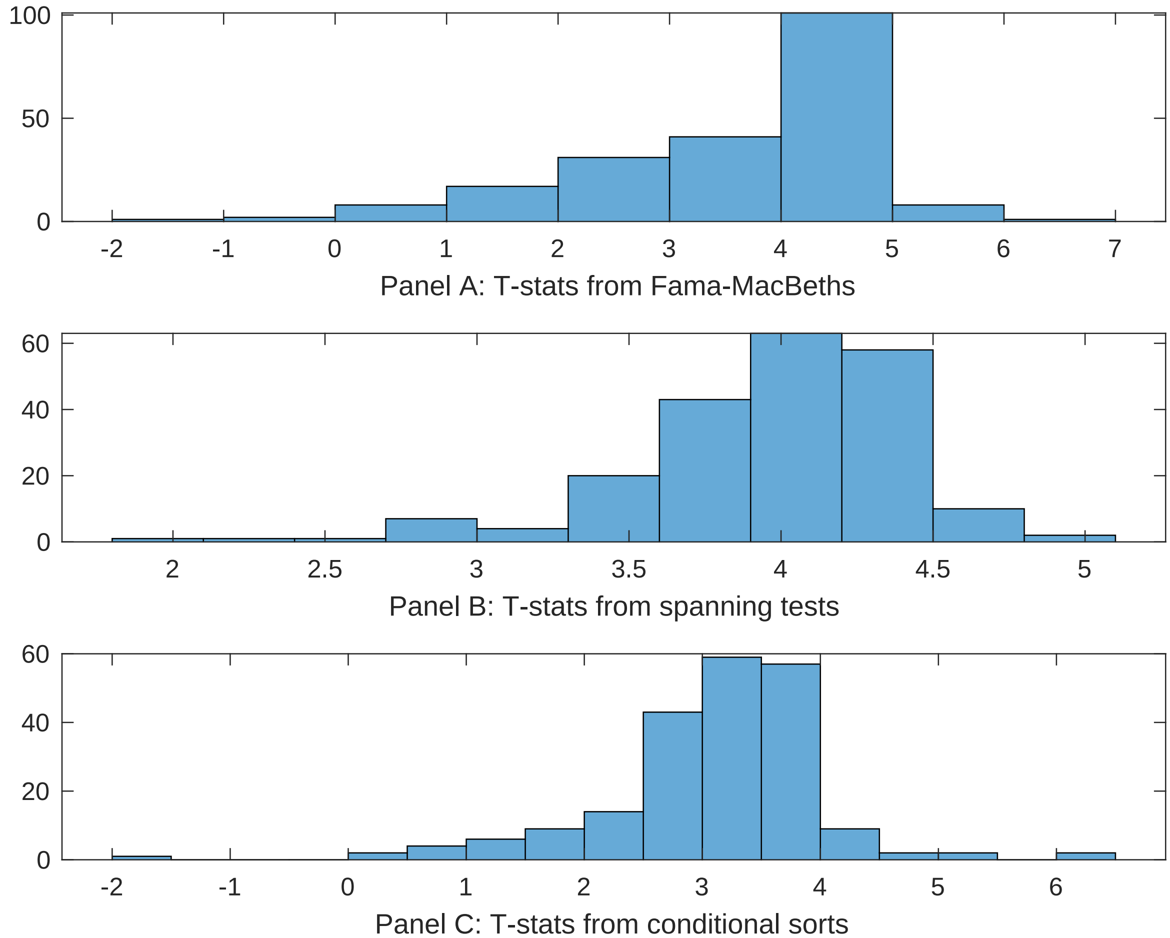


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of EPI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EPI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EPI} EPI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EPI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EPI. Stocks are finally grouped into five EPI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EPI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EPI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EPI} EPI_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13	0.12	0.17	0.13	0.13	0.13	0.14
	[5.98]	[5.43]	[6.64]	[6.37]	[5.61]	[5.42]	[5.00]
EPI	0.74	0.45	0.71	0.69	0.81	0.75	0.44
	[3.87]	[2.07]	[3.32]	[3.66]	[3.92]	[3.47]	[1.90]
Anomaly 1	0.14						0.85
	[6.01]						[3.78]
Anomaly 2		0.21					0.12
		[1.88]					[2.60]
Anomaly 3			0.38				-0.25
			[3.21]				[-0.16]
Anomaly 4				0.27			0.27
				[4.73]			[5.88]
Anomaly 5					0.12		0.98
					[3.17]		[1.63]
Anomaly 6						0.14	-0.15
						[2.45]	[-1.85]
# months	715	715	720	715	720	630	623
$\bar{R}^2$(%)	0	1	0	1	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EPI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EPI} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.18	0.23	0.18	0.12	0.19	0.23	0.19
	[2.33]	[2.97]	[2.35]	[1.63]	[2.48]	[2.84]	[2.43]
Anomaly 1	30.92						13.48
	[8.10]						[2.64]
Anomaly 2		19.34					12.96
		[6.52]					[3.32]
Anomaly 3			33.34				38.69
			[8.18]				[6.22]
Anomaly 4				27.98			14.80
				[6.86]			[2.93]
Anomaly 5					13.83		-18.62
					[3.57]		[-3.26]
Anomaly 6						7.35	-17.37
						[1.96]	[-3.79]
mkt	-0.07	2.16	0.29	0.43	-1.06	-0.41	0.05
	[-0.04]	[1.15]	[0.16]	[0.24]	[-0.57]	[-0.20]	[0.02]
smb	-0.17	1.75	-2.46	0.45	-1.52	5.96	-1.23
	[-0.07]	[0.65]	[-0.95]	[0.17]	[-0.57]	[1.84]	[-0.40]
hml	-1.24	-7.95	-4.43	1.87	-2.15	0.03	-3.81
	[-0.36]	[-2.13]	[-1.26]	[0.53]	[-0.59]	[0.01]	[-1.03]
rmw	-9.97	-4.49	8.35	-3.24	8.04	9.27	2.28
	[-2.45]	[-1.13]	[2.38]	[-0.84]	[2.21]	[2.10]	[0.49]
cma	19.82	22.74	6.73	25.41	25.12	22.65	-6.08
	[3.54]	[3.98]	[1.04]	[4.61]	[3.83]	[3.76]	[-0.88]
umd	2.81	5.76	3.65	2.40	4.59	5.15	3.79
	[1.57]	[3.17]	[2.05]	[1.32]	[2.49]	[2.70]	[2.09]
# months	728	728	732	728	732	630	626
$\bar{R}^2(\%)$	22	20	22	20	16	12	24

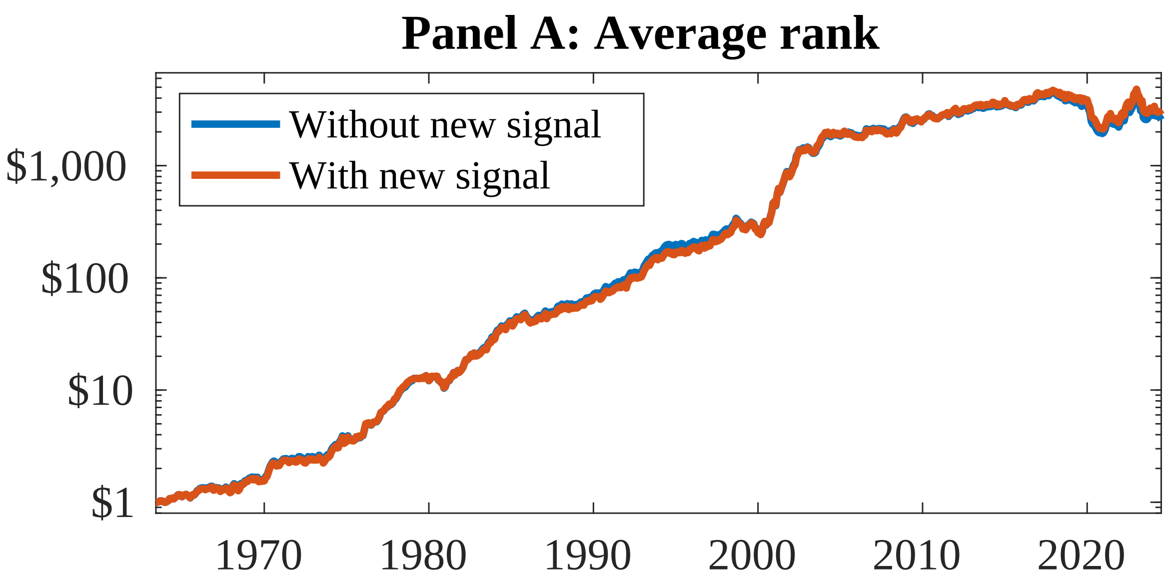


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EPI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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